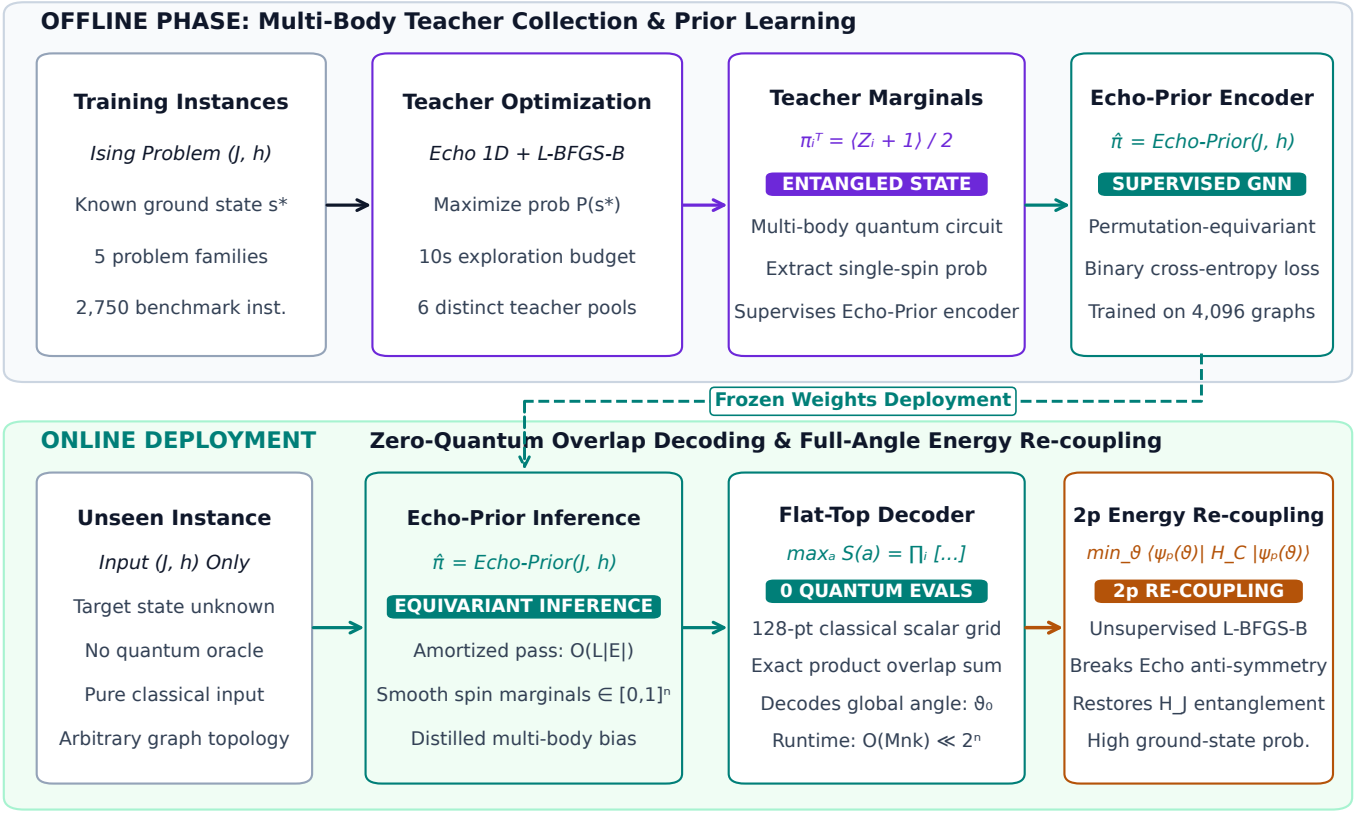
OFFLINE PHASE: Multi-Body Teacher Collection & Prior Learning
Training Instances
Ising Problem (J, h)
Known ground state s*
5 problem families
2,750 benchmark inst.
Teacher Optimization
Echo 1D + L-BFGS-B
Maximize prob P(s*)
10s exploration budget
6 distinct teacher pools
Teacher Marginals
πᵢᵀ = ⟨Zᵢ + 1⟩ / 2
ENTANGLED STATE
Multi-body quantum circuit
Extract single-spin prob
Supervises Echo-Prior encoder
Echo-Prior Encoder
π̂ = Echo-Prior(J, h)
SUPERVISED GNN
Permutation-equivariant
Binary cross-entropy loss
Trained on 4,096 graphs
Frozen Weights Deployment
ONLINE DEPLOYMENT
Zero-Quantum Overlap Decoding & Full-Angle Energy Re-coupling
Unseen Instance
Input (J, h) Only
Target state unknown
No quantum oracle
Pure classical input
Arbitrary graph topology
Echo-Prior Inference
π̂ = Echo-Prior(J, h)
EQUIVARIANT INFERENCE
Amortized pass: O(L|E|)
Smooth spin marginals ∈ [0,1]ⁿ
Distilled multi-body bias
Flat-Top Decoder
maxₐ S(a) = ∏ᵢ [...]
0 QUANTUM EVALS
128-pt classical scalar grid
Exact product overlap sum
Decodes global angle: ϑ₀
Runtime: O(Mnk) ≪ 2ⁿ
2p Energy Re-coupling
min_ϑ ⟨ψₚ(ϑ)| H_C |ψₚ(ϑ)⟩
2p RE-COUPLING
Unsupervised L-BFGS-B
Breaks Echo anti-symmetry
Restores H_J entanglement
High ground-state prob.